

Database Sharding Strategy Report

Patel Ridham Vijaykumar (23110238)
Patel Parthiv Rajeshkumar (23110237)
Laksh Jain (23110185)
Shriniket Behera (23110306)
Rudra Pratap Singh (23110281)

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Abstract

This report details the design and implementation of a database sharding architecture for the IIT-GN Connect web application. It covers the rationale behind the chosen sharding key, the partitioning strategy, the query routing mechanism, and the overall sharding approach. The document also presents a scalability analysis based on benchmarking results and discusses the observed trade-offs and limitations of the system.

Project Links

- **GitHub Repository:** https://github.com/23110281/db_track1_assignment4/
- **Video Demonstration:** <https://youtu.be/ZrMoU3zur8k>

1 Shard Key Selection

1.1 Key Chosen

The `MemberID` was selected as the shard key.

1.2 Justification

The application's data model is centered around the members of the community. Most queries and data relationships are tied directly to a user. For example, posts, poll votes, and group memberships are all associated with a specific `MemberID`.

Choosing `MemberID` as the shard key offers several advantages:

- **Data Colocation:** It allows for the colocation of a member's related data (e.g., their profile, posts, and activities) on a single shard. This significantly improves query performance for most user-centric operations, as it avoids costly cross-shard joins and lookups.
- **Even Data Distribution:** Assuming a uniform distribution of member activity, hashing the `MemberID` helps distribute the data and workload evenly across all shards. This prevents hotspots and ensures that no single shard becomes a bottleneck.
- **Scalability:** As the user base grows, new members can be assigned to shards in a balanced manner, allowing the system to scale horizontally by adding more database shards.

2 Partitioning Strategy

2.1 Strategy Used

A **hash-based partitioning** strategy was implemented. The shard for a given member is determined by the following formula:

$$\text{shard_index} = \text{MemberID} \pmod{N}$$

Where N is the total number of shards (in this case, 3).

2.2 Rationale

Hash partitioning was chosen for its simplicity and effectiveness in achieving uniform data distribution. Unlike range partitioning, which can lead to hotspots if certain ranges of keys are more active than others, hashing a sequential or semi-sequential key like `MemberID` ensures that data is spread pseudo-randomly across the available shards. This approach is ideal for the application's workload, where we want to distribute the load generated by different users as evenly as possible.

3 Query Routing Implementation

Query routing is managed at the application layer within the Flask backend. A dedicated shard routing module, `app/backend/shard_router.py`, centralizes the logic for directing database queries to the appropriate shard.

The core of the routing mechanism is the `get_shard_for_member(member_id)` function, which implements the hash-based partitioning logic.

```
1 def get_shard_for_member(member_id):
2     """Determines the shard index for a given member ID."""
3     shard_index = member_id % 3
4     return shard_index
```

Listing 1: Shard determination logic in `shard_router.py`

API endpoints in the `app/backend/routes/` directory use a set of explicit, shard-aware database functions to perform operations:

- `query_shard(shard_index, query, args)`: Executes a read query on a specific shard.
- `execute_shard(shard_index, query, args)`: Executes a write query on a specific shard.
- `query_all_shards(query, args)`: Executes a read query across all shards and aggregates the results (scatter-gather).
- `execute_all_shards(query, args)`: Executes a write query on all shards (e.g., for replicated tables).
- `insert_replicated_row(...)`: A specialized function to handle inserts into replicated tables, ensuring consistent primary keys across shards.

This application-level routing provides fine-grained control over data access patterns and allows for flexible handling of different query types (e.g., targeting a single shard vs. querying all shards).

4 SQL Shard Tables and Data Migration

4.1 Sharded and Replicated Tables

The database schema was divided into three categories:

1. **Sharded Tables:** Tables containing member-specific data, partitioned by `MemberID`.
 - `Member`
 - `Post`
 - `PostComment`
 - `PostLike`
 - `PollVote`
 - `MemberGroup`
 - `Attendance`
2. **Replicated Tables:** Tables containing global or low-volume data that is replicated across all shards to facilitate joins and avoid cross-shard lookups.
 - `CampusGroup`

- Poll
- JobPost

3. **Local/Control-Plane Tables:** Tables used for application-level concerns that are not part of the sharded data model. These reside in a separate, non-sharded "control" database.

- AuditLog
- OTPVerification

4.2 Data Migration

Data was migrated from a monolithic database structure into the sharded environment using the `seed_shards.py` script. This script reads data from CSV files, determines the correct target shard for each row based on its `MemberID` (for sharded tables), and inserts the data accordingly. For replicated tables, the script iterates through all shards and inserts the data into each one.

5 Sharding Approach and Isolation

5.1 Approach Used

A hybrid approach was adopted. The three primary data shards (`shard0`, `shard1`, `shard2`) are configured as separate MySQL database instances, intended to be run on distinct servers for full physical isolation.

For local development and management of non-sharded data, a `docker-compose.yml` file is used to spin up a single "control-plane" database (`control.db`). This database hosts tables like `AuditLog` that are not part of the core sharded data but are required for the application to function.

5.2 Shard Isolation

Shard isolation is achieved by connecting to each database shard via a unique connection string (host, port, user, password). The application maintains a separate connection pool for each shard. All communication is directed through the application's data access layer, which explicitly routes queries to the correct shard's connection. This ensures that there is no direct communication or data leakage between shards; they operate as independent database servers, with the application acting as the coordinating router.

6 Implementation Journey: A Subtask Report

This section provides a subtask-by-subtask breakdown of the key engineering efforts undertaken to refactor the application from a monolithic proof-of-concept to a compliant, explicitly sharded system.

6.1 Subtask 1: Gap Analysis and Compliance Audit

The first step was a thorough audit of the codebase against the project's documentation. This revealed several critical discrepancies:

- API routes were using generic, non-shard-aware database helpers.
- Inserts into replicated tables returned an incorrect primary key of '0'.
- User registration did not generate sequential 'MemberID's as described.
- The 'docker-compose.yml' file defined a single database, contradicting the multi-shard narrative.

6.2 Subtask 2: Correcting Replicated Table Inserts

Problem: Inserts into replicated tables (e.g., `CampusGroup`) were executed across all shards, but the application layer was receiving a `lastrowid` of `'0'`, causing errors in downstream logic that expected a valid new ID.

Solution: A new function, `insert_replicated_row`, was created. This function performs a canonical insert on the first shard (`shard0`) to obtain a valid auto-incremented ID. It then uses this ID to replay the insert operation on the remaining shards, ensuring data and primary key consistency.

```
1 def insert_replicated_row(table, columns, values):
2     """
3     Inserts a row into a replicated table across all shards,
4     ensuring a consistent primary key.
5     """
6     # 1. Perform canonical insert on shard 0 to get the new ID
7     cols_str = ', '.join(columns)
8     placeholders = ', '.join(['%s'] * len(values))
9     query = f"INSERT INTO {table} ({cols_str}) VALUES ({placeholders})"
10
11     conn_shard0 = get_shard_db_connection(0)
12     cursor_shard0 = conn_shard0.cursor()
13     try:
14         cursor_shard0.execute(query, values)
15         new_id = cursor_shard0.lastrowid
16         conn_shard0.commit()
17     finally:
18         cursor_shard0.close()
19         conn_shard0.close()
20
21     # 2. Replay the insert on other shards with the obtained ID
22     id_column = REPLICATED_ID_COLUMNS.get(table)
23     if not id_column:
24         raise ValueError(f"Unknown ID column for replicated table {table}")
25
26     replay_columns = [id_column] + columns
27     replay_values = [new_id] + list(values)
28     replay_cols_str = ', '.join(replay_columns)
29     replay_placeholders = ', '.join(['%s'] * len(replay_values))
30     replay_query = f"INSERT INTO {table} ({replay_cols_str}) VALUES ({replay_placeholders})"
31
32     for i in range(1, 3): # Shards 1 and 2
33         conn_shard_i = get_shard_db_connection(i)
34         cursor_shard_i = conn_shard_i.cursor()
35         try:
36             cursor_shard_i.execute(replay_query, replay_values)
37             conn_shard_i.commit()
38         finally:
39             cursor_shard_i.close()
40             conn_shard_i.close()
41
42     return new_id
```

Listing 2: New replicated insert logic in `shard_db.py`

6.3 Subtask 3: Refactoring API Routes to be Shard-Explicit

Problem: API endpoints were using legacy helpers (`query_db`, `execute_db`) that obfuscated the shard-ing logic, violating the principle of explicit, endpoint-level routing.

Solution: All API routes in the `app/backend/routes/` directory were systematically refactored. Calls to the legacy helpers were replaced with explicit calls to the new shard-aware functions: `query_shard`, `execute_shard`, `query_all_shards`, and the new `insert_replicated_row`.

```
1 # Old implementation
2 @groups_bp.route('/', methods=['POST'])
3 @jwt_required()
4 def create_group():
5     # ...
6     group_id = execute_db(
7         "INSERT INTO CampusGroup (Name, Description) VALUES (%s, %s)",
8         (name, description))
```

```

9     )
10    # ...

```

Listing 3: Before: Implicit routing in routes/groups.py

```

1  # New implementation
2  from app.backend.shard_db import insert_replicated_row
3
4  @groups_bp.route('/', methods=['POST'])
5  @jwt_required()
6  def create_group():
7      # ...
8      group_id = insert_replicated_row(
9          'CampusGroup',
10         ['Name', 'Description'],
11         [name, description]
12     )
13     # ...

```

Listing 4: After: Explicit routing in routes/groups.py

6.4 Subtask 4: Implementing Sequential MemberID Generation

Problem: New users were being assigned a random MemberID, which did not align with the documentation’s narrative of a sequential, growing user base.

Solution: The user registration logic in `app/backend/routes/auth.py` was modified. It now performs a scatter-gather query to find the maximum MemberID across all three shards before inserting a new member, ensuring the new MemberID is sequential.

```

1  # Get the maximum MemberID across all shards
2  max_id_results = query_all_shards("SELECT MAX(MemberID) as max_id FROM Member", ())
3  max_id = 0
4  for result in max_id_results:
5      if result and result['max_id'] is not None:
6          max_id = max(max_id, result['max_id'])
7  new_member_id = max_id + 1
8
9  # Determine the correct shard for the new member
10 shard_index = get_shard_for_member(new_member_id)
11
12 # Insert the new member into the designated shard
13 execute_shard(
14     shard_index,
15     "INSERT INTO Member (MemberID, Name, Email, Password) VALUES (%s, %s, %s, %s)",
16     (new_member_id, name, email, hashed_password)
17 )

```

Listing 5: Sequential ID logic in routes/auth.py

7 Scalability and Trade-offs Analysis

7.1 Benchmark Results

The system was benchmarked using a script (`benchmark.py`) that simulates read and write workloads. The benchmarks were run on both the original, non-sharded (monolithic) database and the new sharded architecture.

Note: The following is a summary based on the project’s benchmark results. Please replace with your specific findings from `benchmark_results.json`.

The results indicate a significant improvement in write throughput for the sharded system. Because write operations on sharded tables are directed to a specific shard, the system can handle approximately three times the write load, as the work is distributed.

Read performance for single-shard queries (e.g., fetching a user’s profile) remained comparable to the monolithic setup, as expected. However, for scatter-gather queries that hit all shards (e.g., aggregating all posts), there was a noticeable latency increase due to the overhead of connecting to multiple databases and merging results in the application.

7.2 Trade-offs

- **Increased Complexity:** The sharded architecture is significantly more complex than the monolithic version. It requires careful management of database connections, sophisticated routing logic, and a strategy for handling replicated data.
- **Join Limitations:** Performing joins across shards is not feasible. This limitation was mitigated by replicating smaller, frequently accessed tables (`CampusGroup`, `Poll`, `JobPost`).
- **Transactional Integrity:** Ensuring transactional integrity across multiple shards is challenging. This implementation restricts transactions to a single shard to maintain simplicity and consistency.
- **Operational Overhead:** Managing three separate database instances (plus a control database) increases operational overhead for backups, monitoring, and maintenance.

8 Observations and Limitations

- **Sequential ID Generation:** A key challenge was generating unique, sequential IDs for new members. The solution involves a `'MAX(MemberID)'` query across all shards before an insert, which introduces a small amount of latency and a potential race condition under very high concurrency. A more robust solution might involve a dedicated ID generation service (like Twitter's Snowflake).
- **Replicated Table Consistency:** The `insert_replicated_row` function ensures that new rows in replicated tables have consistent primary keys. However, updates and deletes to these tables must be carefully propagated to all shards to maintain consistency.
- **Hotspots:** While hash partitioning on `MemberID` generally provides even distribution, it's possible for a "celebrity" user (one with vastly more activity and data) to create a hotspot on their assigned shard.
- **Rebalancing:** The current implementation does not include a mechanism for rebalancing data if a shard becomes full or if more shards are added to the system. Rebalancing is a complex operation that would require a carefully planned data migration strategy.